

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score
)

# Load Dataset
df = pd.read_csv("Housing.csv")

# Display Dataset
print(df.head())

# Check Missing Values
print(df.isnull().sum())

# Fill Missing Values
df.fillna(df.mean(numeric_only=True), inplace=True)

# Features
X = df[['area', 'bedrooms', 'bathrooms', 'stories', 'parking']]

# Target Variable
y = df['price']

# Train Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Create Model
model = LinearRegression()

# Train Model
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Evaluation Metrics
mae = mean_absolute_error(y_test, y_pred)

mse = mean_squared_error(y_test, y_pred)

r2 = r2_score(y_test, y_pred)

print("\nModel Evaluation")

print("MAE :", mae)

print("MSE :", mse)

print("R2 Score :", r2)

# Coefficients
print("\nIntercept:")
print(model.intercept_)

print("\nCoefficients:")

for feature, coef in zip(X.columns, model.coef_):
    print(feature, ":", coef)
```

```
# Actual vs Predicted
result = pd.DataFrame({
    "Actual": y_test,
    "Predicted": y_pred
})

print(result.head())

# Plot Actual vs Predicted
plt.figure(figsize=(8,5))

plt.scatter(y_test, y_pred)

plt.xlabel("Actual Price")

plt.ylabel("Predicted Price")

plt.title("Actual vs Predicted House Prices")

plt.show()
```

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	\
0	13300000	7420	4	2	3	yes	no	no	
1	12250000	8960	4	4	4	yes	no	no	
2	12250000	9960	3	2	2	yes	no	yes	
3	12215000	7500	4	2	2	yes	no	yes	
4	11410000	7420	4	1	2	yes	yes	yes	

	hotwaterheating	airconditioning	parking	prefarea	furnishingstatus
0	no	yes	2	yes	furnished
1	no	yes	3	no	furnished
2	no	no	2	yes	semi-furnished
3	no	yes	3	yes	furnished
4	no	yes	2	no	furnished

```

price      0
area       0
bedrooms   0
bathrooms  0
stories    0
mainroad   0
guestroom  0
basement   0
hotwaterheating  0
airconditioning  0
parking       0
prefarea      0
furnishingstatus  0
dtype: int64

```

Model Evaluation

MAE : 1127483.352323519
MSE : 2292721545725.3623
R2 Score : 0.5464062355495871

Intercept:

51999.67680883873

Coefficients:

area : 308.86695608764126
bedrooms : 151246.75062952132
bathrooms : 1185731.7137011932
stories : 495100.7626617581
parking : 337660.8302982585
Actual Predicted